to think seriously about his PhD thesis. Everett wanted to do a short and fun thesis, but he needed a suitable subject. It occurred to him over drinks. “One night at the Graduate College after a slosh or two of sherry,” Everett recalled, talking with Misner years later, “you and Aage were starting to say some ridiculous things about the implications of quantum mechanics and I was having a little fun, joshing you and telling you some of the outrageous implications of what you said, and, ah, as we had a little more sherry and got a little more potted in the conversation—don’t you remember, Charlie? You were there!” Misner didn’t remember, which Everett chalked up to “too much sherry,” and continued:

> *Everett:* Well, anyway, the whole business started with those discussions, and my impression is I went to Wheeler then later and said, “Hey, how about this, this is the thing to do.”… [T]his obvious inconsistency in the [quantum] theory or whatever I thought of it then.…
>
> *Misner:* It is strange that he would be so interested in it—all in all, because it certainly went against the normal tenets of his great master, Bohr.
>
> *Everett:* Well, he still feels that way a little bit.

At the time, Wheeler “was preaching this idea that you ought to just look at the equations and obey the fundamentals of physics while you follow their conclusions and give them a serious hearing,” according to Misner. For his PhD thesis, Everett took Wheeler’s advice. He looked at the outrageous implications of quantum physics and gave them a serious hearing. What he found was far more astonishing than anything in his beloved science-fiction stories.

We met the measurement problem back in Chapter 1. The problem, in a nutshell, is this: Quantum wave functions move along nice and smoothly, always obeying one simple and deterministic law, the Schrödinger equation—except when they don’t. When a measurement happens, wave functions collapse. How and why wave function collapse happens—and what constitutes a “measurement” anyway—is the measurement problem, the